Sketching Rational Functions

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Sketching Rational Functions

Teaching Notes, Worked Examples and Practice

1. What is a rational function?

A **rational function** is a function that can be written as a **fraction of two polynomials**:

$$f(x) = \frac{P(x)}{Q(x)}, \qquad Q(x) \neq 0$$

The word *rational* comes from *ratio* – just like a rational number $\frac{p}{q}$ is a ratio of two integers.

Our focus: nonconstant denominators.

We can never divide by zero, so any x-value that makes $Q(x) = 0$ is **not allowed**. This is why the graph of a rational function **breaks into separate pieces** (branches).

Function	Rational?	Reason
$f(x) = \frac{1}{x}$	✓	1 and x are both polynomials
$f(x) = \frac{2x+4}{x+1}$	✓	linear ÷ linear
$f(x) = \frac{9}{x+2} + 3$	✓	can be written as $\frac{3x+15}{x+2}$
$f(x) = 3x^2 - 5$	Yes	A polynomial is a ratio with denominator 1.
$f(x) = \frac{1}{\sqrt{x}}$	×	$\sqrt{x}$ is not a polynomial

2. Rational functions in the IB AA syllabus

Level	Form	What the graph has
SL & HL	$f(x) = \frac{1}{x}$ and $f(x) = \frac{ax+b}{cx+d}$	1 vertical + 1 horizontal asymptote
HL only	$f(x) = \frac{ax+b}{cx^2+dx+e}$	up to 2 vertical asymptotes, H.A. $y = 0$
HL only	$f(x) = \frac{ax^2+bx+c}{dx+e}$	1 vertical + 1 **oblique** asymptote

3. Key words

- **Asymptote:** a line the graph gets closer and closer to but never reaches (at the ends of the graph).

Vertical asymptote: a non-removable denominator zero.

Key notes

- **Horizontal asymptote (H.A.):** the value y approaches as $x \to \pm\infty$. Equation $y = k$.
- **Oblique (slant) asymptote:** a sloping line $y = mx + c$ that the graph approaches as $x \to \pm\infty$.
- **Intercepts:** where the graph crosses the axes. **Domain / range:** allowed x-values / y-values.

Always write asymptotes as equations, e.g. "$x = -2$" and "$y = 3$", not just "-2" and "3".

4. Steps to sketch a rational function

Step	What to do	How
1	Factorise and simplify	factorise the top and the bottom
2	x-intercept(s)	put the **numerator** $= 0$ (i.e. $y = 0$)
3	y-intercept	put $x = 0$
4	Vertical asymptote(s)	put the **denominator** $= 0$
5	Horizontal / oblique asymptote	compare degrees (see the table below)
6	Domain	$x \neq$ any value that makes the denominator 0
7	Sign diagram	where the graph is above $(+)$ or below $(-)$ the x-axis
8	Draw the graph	asymptotes first (dashed), then points, then the branches
9	Range	read it from your finished graph

5. Finding the horizontal (or oblique) asymptote – compare degrees

Degree = highest power of x. Let N = degree of numerator, D = degree of denominator.

Case	Example	Asymptote
N < D	$\dfrac{x+1}{x^2-x-6}$	H.A. is $y = 0$ (the x-axis)
N = D	$\dfrac{2x+4}{x+1}$	H.A. is $y = \dfrac{\text{coefficient of } x \text{ on top}}{\text{coefficient of } x \text{ on bottom}} = \dfrac{2}{1} = 2$
N = D +1	$\dfrac{x^2-x}{x+1}$	**no** H.A. – an **oblique** asymptote; divide to find it

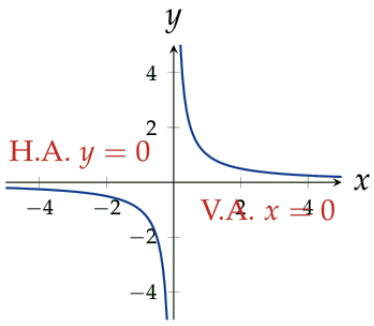
Shortcut for the form $y = a + \dfrac{b}{x-h}$: V.A. $x = h$, H.A. $y = a$. e.g. $y = \dfrac{9}{x+2} + 3$: V.A. $x = -2$, H.A. $y = 3$.

6. Domain and range

Domain: exclude every zero of the original denominator.

- **Range** for $y = \dfrac{ax+b}{cx+d}$ (SL): all real y except the H.A.: $y \neq$ H.A.
- **Range** for HL forms: look at the graph – use turning points from the GDC (the graph may cross $y = 0$).

7. The basic graph $y = \dfrac{1}{x}$ and transformations



The graph of $y = \dfrac{1}{x}$ is a **rectangular hyperbola** with asymptotes $x = 0$ and $y = 0$.

$y = a + \dfrac{b}{x-h}$ is $y = \dfrac{1}{x}$ after:

- a vertical stretch, scale factor b
- a translation h units right
- a translation a units up

So the asymptotes move to $x = h$ and $y = a$.

8. How to make a sign diagram

A sign diagram tells you where the graph is **above** the x-axis $(+)$ and **below** it $(-)$.

1. Write all the x-intercepts and vertical asymptotes in order along the top.
2. Make one row for each factor (top and bottom).
3. For an $(x+a)$ factor: **negative to the left** of its value, **positive to the right**.
   For an $(a-x)$ factor the signs are the **opposite**: positive to the left, negative to the right.
   A squared factor, or one like x^2+4 that is never zero, is **always positive**.
4. The last row (total) is found by **multiplying the signs** in each column.

Example: $f(x) = \dfrac{x+1}{(x+2)(x-3)}$ (0 = x-intercept, $\|$ = vertical asymptote)

	-2		-1		3		
	$-$		$-$	0	$+$	$+$	$(x+1)$
	$-$	0	$+$		$+$	$+$	$(x+2)$
	$-$		$-$		$-$	0 $+$	$(x-3)$
	$-$	$\|$	$+$	0	$-$	$\|$ $+$	$f(x)$

So the graph is **below** the x-axis for $x < -2$ and $-1 < x < 3$, and **above** it for $-2 < x < -1$ and $x > 3$.

9. Finding an oblique asymptote (HL) – divide

When the top has degree one more than the bottom, divide (long division or synthetic

division):

$$\frac{2x^2-5x+6}{x-3} \qquad \begin{array}{c|ccc} 3 & 2 & -5 & 6 \\ & & 6 & 3 \\ \hline & 2 & 1 & \boxed{9} \end{array} \qquad \Rightarrow \qquad \frac{2x^2-5x+6}{x-3} = 2x+1+\frac{9}{x-3}$$

As $x \to \pm\infty$, $\dfrac{9}{x-3} \to 0$, so the graph approaches the line $\boxed{y = 2x+1}$. The remainder is ignored for the asymptote.

10. Common mistakes

- Writing the asymptote as a number instead of an equation ($x = 3$, not "3").
- Letting the curve **touch or cross a vertical asymptote** – it never can.
- Forgetting to label intercepts with coordinates, e.g. $(0, -2)$.
- Drawing branches in the wrong "corners" – always check with the sign diagram or one test point.
- Assuming a graph can never cross a *horizontal* asymptote – in the middle of the graph it can (e.g. Worked example 4).

Checked conventions

Rational functions. Every polynomial is also rational: write it over denominator 1. This booklet focuses mainly on nonconstant denominators; the source's polynomial classification has been corrected in the notes table.

Domain and holes. Exclude every zero of the original denominator. Cancelling a common factor creates a hole; it does not restore the excluded x-value. A vertical asymptote occurs only at a non-removable denominator zero.

Range. For a nonconstant linear-over-linear function with a genuine vertical asymptote, the excluded horizontal-asymptote height is its only excluded range value. For other rational forms, use turning points, limits and the graph; a horizontal asymptote can be crossed.

Inverse graphs. A function and its inverse are reflections in $y = x$, but their intersections are not universally restricted to that line. Solve the actual equations and apply their domains.

Correct argument for Q20(b). With $f(x) = \frac{a-3x}{x-2}$, its inverse is $f^{-1}(x) = \frac{a+2x}{x+3}$ when $a \neq 6$. The given equality gives $3 - a = \frac{a+2}{4}$, so $12 - 4a = a + 2$ and $\boldsymbol{a = 2}$. This replaces the source's unjustified blanket claim about inverse intersections.

Q24(e). Solve $\frac{3x+4}{2x-5} = \frac{5x+4}{2x-3}$ with $x \neq \frac{5}{2}, \frac{3}{2}$. Simplifying gives $x^2 - 4x - 2 = 0$, hence $x = 2 \pm \sqrt{6}$. Substitution confirms the two intersection points are $\left(2 \pm \sqrt{6}, 2 \pm \sqrt{6}\right)$ for this particular function.

Worked examples

Worked example 1 SL & HL

Sketch $y = \dfrac{9}{x+2} + 3$.

V.A.: $x + 2 = 0 \Rightarrow \boxed{x = -2}$

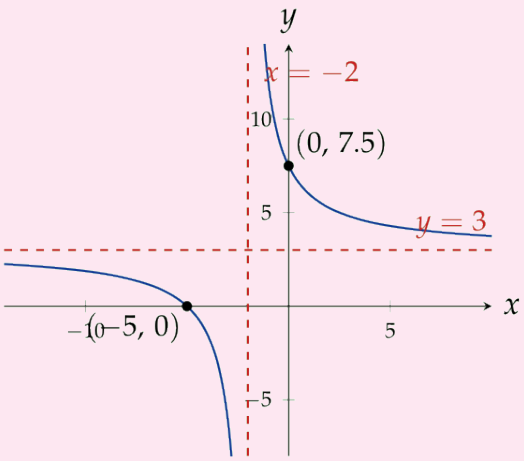
H.A.: $y = 0 + 3 \Rightarrow \boxed{y = 3}$

***y*-int:** $x = 0$: $y = \frac{9}{2} + 3 = 7.5 \Rightarrow (0, 7.5)$

***x*-int:** $0 = \frac{9}{x+2} + 3$

$-3(x + 2) = 9 \Rightarrow x = -5 \Rightarrow (-5, 0)$

Domain: $x \neq -2$ **Range:** $y \neq 3$



Worked example 2 SL & HL

Sketch $y = \dfrac{2x+4}{x+1}$.

V.A.: $x + 1 = 0 \Rightarrow \boxed{x = -1}$

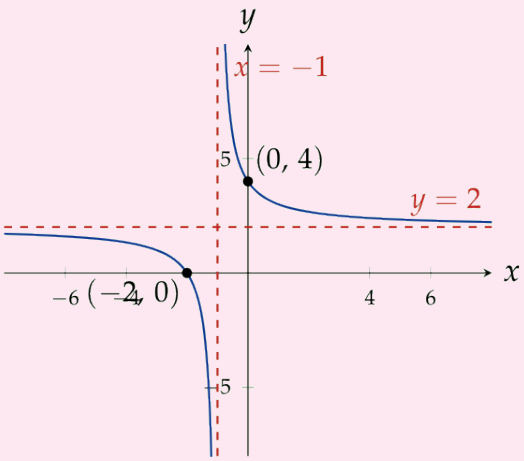
H.A.: same degree: $y = \frac{2}{1} \Rightarrow \boxed{y = 2}$

***y*-int:** $y = \frac{4}{1} = 4 \Rightarrow (0, 4)$

***x*-int:** $2x + 4 = 0 \Rightarrow x = -2 \Rightarrow (-2, 0)$

Domain: $x \neq -1$ **Range:** $y \neq 2$

Check: write it as $y = 2 + \frac{2}{x+1}$ *– the same asymptotes.*



Worked examples

Worked example 3 HL only

Sketch $y = \dfrac{1}{x^2+3x+2}$.

1. Factorise: $y = \dfrac{1}{(x+2)(x+1)}$

2. x-int: numerator is $1 \neq 0$ – **none**

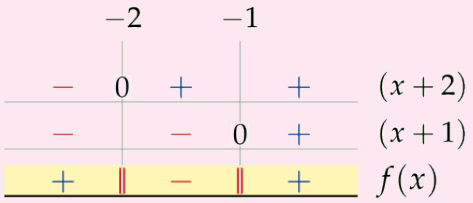
3. y-int: $y = \frac{1}{2} \Rightarrow \left(0, \frac{1}{2}\right)$

4. V.A.: $\boxed{x=-2}$ and $\boxed{x=-1}$

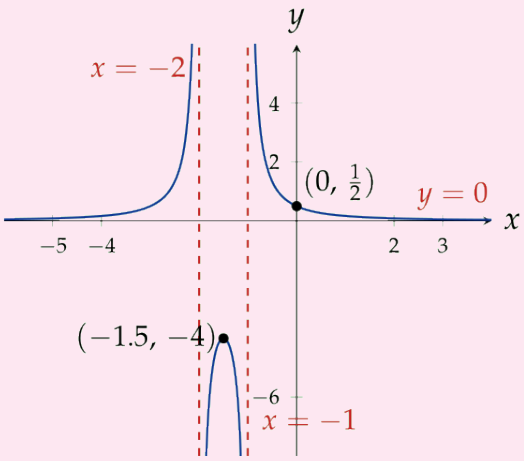
5. H.A.: N<D $\Rightarrow \boxed{y=0}$

6. Domain: $x \neq -2,\ x \neq -1$

7. Sign diagram:



Range: the GDC gives a maximum at $(-1.5, -4)$ between the asymptotes: $y > 0$ or $y \leq -4$.



x −1 −2 −1.5 −2⁻
f(x) 0 0 −¼ 0⁻
1/f(x) α α −4 α⁻

Tip: why is the middle branch ∩-shaped? – the "reciprocal of the parabola" idea

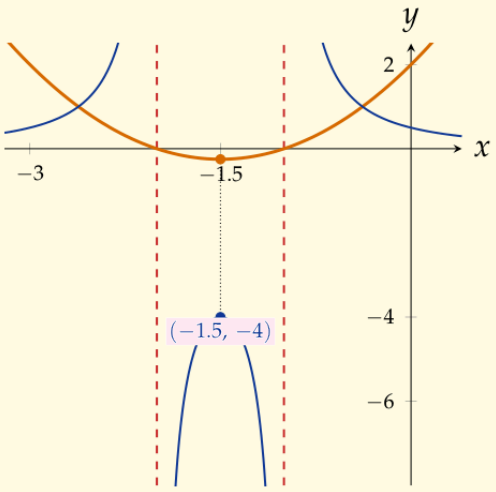
The denominator $q(x) = x^2+3x+2$ is a **U-shaped parabola**. Its minimum is at $(-1.5, -0.25)$, and it is **negative between** -2 **and** -1.

The graph is $y = \dfrac{1}{q}$, so:

- where q is a **small negative** number (near -2 and -1), $\frac{1}{q}$ is a **huge negative** number;
- where q is at its lowest, -0.25, $\frac{1}{q} = \frac{1}{-0.25} = -4$. This is the **least negative** value, so it is the **top of the** ∩.

So the **dip of the U** in the denominator turns into the **hump of the** ∩ in the graph, and the turning point is at the **same x-value**.

This explains why the maximum is at $x = -1.5$ – no calculus or GDC needed.



orange: $q = x^2+3x+2$

blue: $y = \dfrac{1}{q}$

Worked examples

Worked example 4 HL only

Sketch $y = \dfrac{x+1}{x^2-x-6}$.

1. Factorise: $y = \dfrac{x+1}{(x-3)(x+2)}$

2. x-int: $x+1=0 \Rightarrow (-1,\,0)$

3. y-int: $y = \frac{1}{-6} \Rightarrow \left(0,\,-\frac{1}{6}\right)$

4. V.A.: $\boxed{x=3}$ and $\boxed{x=-2}$

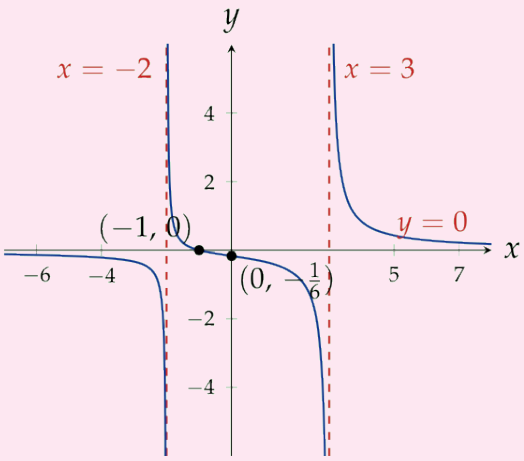
5. H.A.: N<D $\Rightarrow$ $\boxed{y=0}$

6. Domain: $x \neq 3,\ x \neq -2$

7. Signs (from section 8): $-\quad +\quad -\quad +$

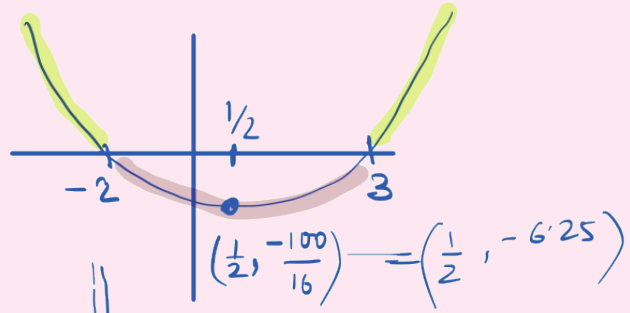
8. Graph: the middle branch goes from $+\infty$ down through $(-1,0)$ to $-\infty$ – it **crosses** the H.A. $y=0$ at the x-intercept.

Range: $y \in \mathbb{R}$ (every value is reached by the middle branch).



$x^2 - x - 6$ $\quad (x-3)(x+2)$

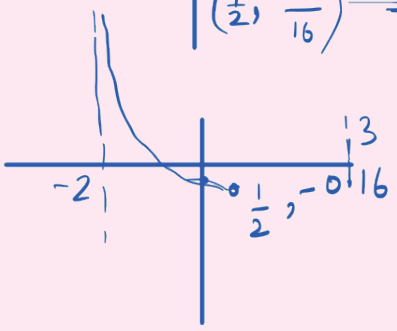
$x = 3 \qquad x = -2$



$-2.5\ (2.5)$

$-\frac{10}{4}\ \left|\ \frac{10}{4}\right.$

$-\frac{100}{16}$



Worked examples

Worked example 5 HL only

Sketch $f(x) = \dfrac{2x^2 - 5x + 6}{x - 3}$.

V.A.: $\boxed{x = 3}$

Oblique asymptote: N = D +1, so divide (section 9):

$f(x) = 2x + 1 + \dfrac{9}{x-3} \Rightarrow \boxed{y = 2x + 1}$

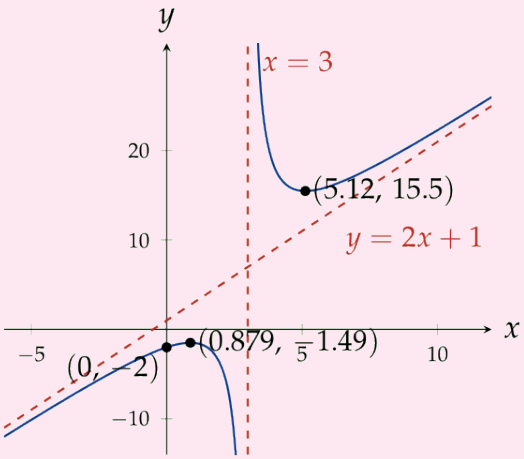
x-int: $2x^2 - 5x + 6 = 0$, $\Delta = 25 - 48 < 0$ – **none**

y-int: $f(0) = \frac{6}{-3} = -2 \Rightarrow (0, -2)$

Turning points (GDC):
max $(0.879, -1.49)$, min $(5.12, 15.5)$

Domain: $x \neq 3$

Range: $y \leq -1.49$ or $y \geq 15.5$



Worked example 6 HL only

Sketch $y = \dfrac{x^2 - x}{x + 1}$.

Factorise: $y = \dfrac{x(x-1)}{x+1}$

x-int: $(0, 0)$ and $(1, 0)$; **y-int:** $(0, 0)$

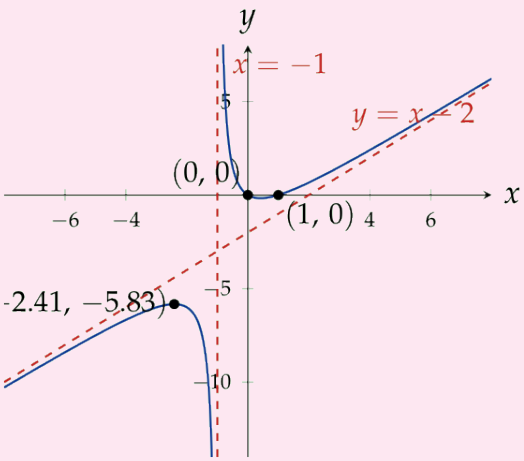
V.A.: $\boxed{x = -1}$

Oblique asymptote: divide by $(x + 1)$:

$$\begin{array}{c|ccc} -1 & 1 & -1 & 0 \\ & & -1 & 2 \\ \hline & 1 & -2 & \boxed{2} \end{array} \Rightarrow y = x - 2 + \frac{2}{x+1}$$

so the oblique asymptote is $\boxed{y = x - 2}$.

Turning points (GDC): max $(-2.41, -5.83)$, min $(0.414, -0.172)$



Question 1

Source worksheet · p.9

Question 1

Non-calculator SL & HL Easy [Maximum mark: 6]

Write down the equations of the vertical and horizontal asymptotes of:

(a) $y = \frac{3}{x-4}$ (b) $y = \frac{2}{x+1} + 5$ (c) $y = \frac{x+3}{x-2}$

(d) $y = \frac{4x-1}{2x+6}$ (e) $y = \frac{6-3x}{x+5}$ (f) $y = \frac{5}{2x-1} - 3$

Working:

Question 2

Source worksheet · p.9

Question 2

Non-calculator SL & HL Easy [Maximum mark: 6]

Find the coordinates of the x- and y-intercepts (if they exist) of:

(a) $y = \frac{x+3}{x-2}$ (b) $y = \frac{4x-1}{2x+6}$ (c) $y = \frac{2}{x+1} + 5$ (d) $y = \frac{3}{x-4}$

Working:

Answer 1 Source worksheet · p.9

Question 1

	Function	V.A.	H.A.
(a)	$\dfrac{3}{x-4}$	$x=4$	$y=0$ (number on top, x only below)
(b)	$\dfrac{2}{x+1}+5$	$x=-1$	$y=5$ (the "+5")
(c)	$\dfrac{x+3}{x-2}$	$x=2$	$y=\frac{1}{1}=1$
(d)	$\dfrac{4x-1}{2x+6}$	$2x+6=0\Rightarrow x=-3$	$y=\frac{4}{2}=2$
(e)	$\dfrac{6-3x}{x+5}$	$x=-5$	$y=\frac{-3}{1}=-3$ (use the coefficient of x, with its sign)
(f)	$\dfrac{5}{2x-1}-3$	$2x-1=0\Rightarrow x=\frac{1}{2}$	$y=-3$

Answer 2 Source worksheet · p.9

Question 2

(a) x-int: $x+3=0\Rightarrow \boxed{(-3,0)}$; y-int: $y=\frac{3}{-2}\Rightarrow \boxed{\left(0,-\frac{3}{2}\right)}$

(b) x-int: $4x-1=0\Rightarrow \boxed{\left(\frac{1}{4},0\right)}$; y-int: $y=\frac{-1}{6}\Rightarrow \boxed{\left(0,-\frac{1}{6}\right)}$

(c) y-int: $y=\frac{2}{1}+5=7\Rightarrow \boxed{(0,7)}$;

x-int: $\frac{2}{x+1}=-5\Rightarrow x+1=-\frac{2}{5}\Rightarrow x=-\frac{7}{5}\Rightarrow \boxed{\left(-\frac{7}{5},0\right)}$

(d) x-int: **none** (the numerator 3 is never 0); y-int: $y=\frac{3}{-4}\Rightarrow \boxed{\left(0,-\frac{3}{4}\right)}$

Question 3 Source worksheet · p.11

Question 3 Non-calculator SL & HL Easy [Maximum mark: 4]

Write down the domain and range of:

(a) $f(x) = \dfrac{x+3}{x-2}$ (b) $f(x) = \dfrac{2}{x+1} + 5$ (c) $f(x) = \dfrac{6-3x}{x+5}$ (d) $f(x) = \dfrac{3}{x-4}$

Working:

Question 4 Source worksheet · p.11

Question 4 Non-calculator SL & HL Medium [Maximum mark: 6]

Let $f(x) = \dfrac{2x+5}{x+1}$.

(a) Write $f(x)$ in the form $a + \dfrac{b}{x+1}$.

(b) Hence write down the equations of the asymptotes.

(c) Describe a sequence of transformations that maps the graph of $y = \dfrac{1}{x}$ onto the graph of $y = f(x)$.

Working:

Answer 3 Source worksheet · p.11

Question 3

Domain: $x \neq$ V.A. Range: $y \neq$ H.A.

	Function	Domain	Range
(a)	$\dfrac{x+3}{x-2}$	$x \neq 2$	$y \neq 1$
(b)	$\dfrac{2}{x+1}+5$	$x \neq -1$	$y \neq 5$
(c)	$\dfrac{6-3x}{x+5}$	$x \neq -5$	$y \neq -3$
(d)	$\dfrac{3}{x-4}$	$x \neq 4$	$y \neq 0$

Answer 4 Source worksheet · p.11

Question 4

(a) $\dfrac{2x+5}{x+1} = \dfrac{2(x+1)+3}{x+1} = \boxed{2+\dfrac{3}{x+1}}$ $(a=2,\ b=3)$

(b) V.A.: $\boxed{x=-1}$ H.A.: $\boxed{y=2}$

(c) $\dfrac{1}{x} \to \dfrac{3}{x} \to \dfrac{3}{x+1} \to \dfrac{3}{x+1}+2$

- vertical stretch, scale factor 3
- translation 1 unit to the **left**
- translation 2 units **up** (i.e. translation by $\binom{-1}{2}$ after the stretch)

Question 5

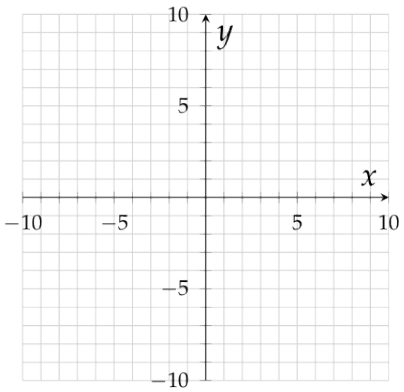
Source worksheet · p.13

Question 5

Non-calculator SL & HL Easy [Maximum mark: 6]

Sketch $f(x) = \dfrac{4}{x-1} - 2$, and state the domain and range.

Working:



Question 6

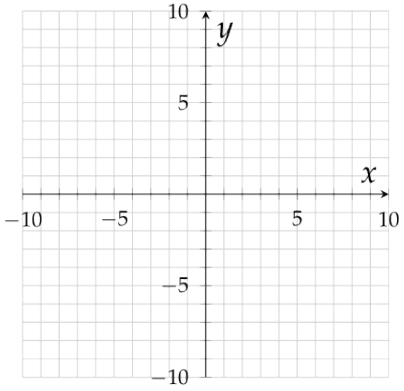
Source worksheet · p.13

Question 6

Non-calculator SL & HL Easy [Maximum mark: 6]

Sketch $f(x) = \dfrac{3x-6}{x+3}$, and state the domain and range.

Working:



Answer 5 Source worksheet · p.13

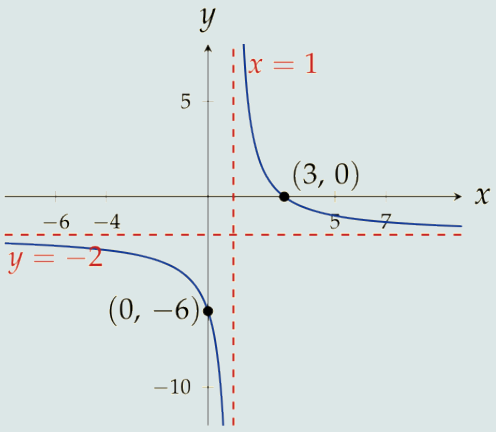
Question 5

V.A.: $\boxed{x = 1}$ H.A.: $\boxed{y = -2}$

y-int: $f(0) = \frac{4}{-1} - 2 = -6 \Rightarrow (0, -6)$

x-int: $\frac{4}{x-1} = 2 \Rightarrow x - 1 = 2 \Rightarrow (3, 0)$

Domain: $x \neq 1$ Range: $y \neq -2$



Answer 6 Source worksheet · p.13

Question 6

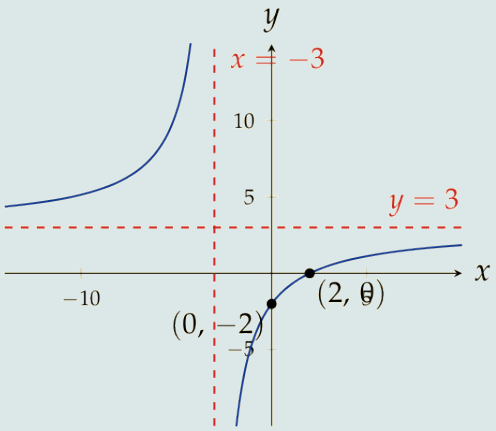
V.A.: $x + 3 = 0 \Rightarrow \boxed{x = -3}$

H.A.: $y = \frac{3}{1} \Rightarrow \boxed{y = 3}$

x-int: $3x - 6 = 0 \Rightarrow (2, 0)$

y-int: $\frac{-6}{3} = -2 \Rightarrow (0, -2)$

Domain: $x \neq -3$ Range: $y \neq 3$

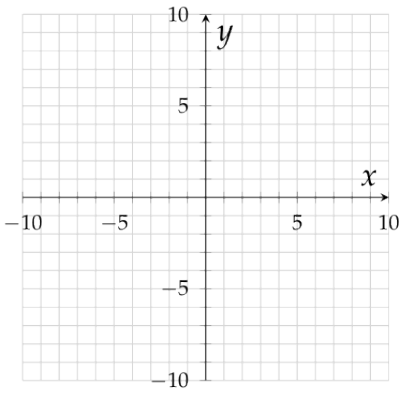


Question 7 Source worksheet · p.15

Question 7 Non-calculator SL & HL Medium [Maximum mark: 6]

Sketch $f(x) = \dfrac{5-2x}{x+4}$, and state the domain and range.

Working:



Question 18 Source worksheet · p.27

Question 18 Non-calculator SL & HL Easy [Maximum mark: 8]

Source: extra practice question

Let $h(x) = \dfrac{2}{x+1} - 3,\ x \neq -1$.

(a) Write down the equations of the vertical and horizontal asymptotes. [2]

(b) Find the coordinates of the points where the graph of h meets the axes. [3]

(c) Sketch the graph of h, showing the asymptotes and intercepts. [3]

Working:

Answer 7 Source worksheet · p.15

Question 7

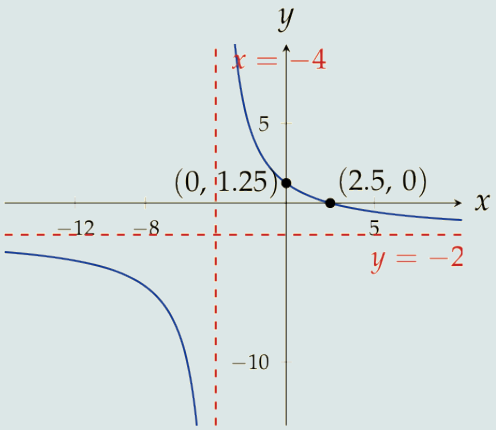
V.A.: $\boxed{x = -4}$

H.A.: $y = \frac{-2}{1} \Rightarrow \boxed{y = -2}$ (careful with the sign of $-2x$)

x-int: $5 - 2x = 0 \Rightarrow (2.5,\ 0)$

y-int: $\frac{5}{4} \Rightarrow (0,\ 1.25)$

Domain: $x \neq -4$ Range: $y \neq -2$



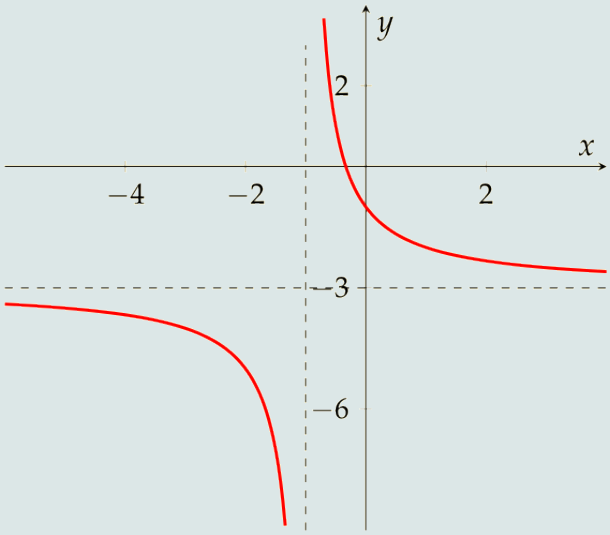
Answer 18 Source worksheet · p.27

Question 18

(a) $\boxed{x = -1}$, $\boxed{y = -3}$

(b) y-int: $h(0) = 2 - 3$: $\boxed{(0,\ -1)}$; x-int: $\dfrac{2}{x+1} = 3 \Rightarrow x + 1 = \frac{2}{3}$: $\boxed{\left(-\frac{1}{3},\ 0\right)}$

(c)



Question 19 Source worksheet · p.27

Question 19 Non-calculator SL & HL Easy [Maximum mark: 8]

Source: Revision Village (Rational & Inverse Functions Q7)

Let $f(x) = \dfrac{3x-6}{x+1}$, for $x \neq -1$.

(a) For the graph of $y = f(x)$, find the coordinates of:

(i) the x-intercept;

(ii) the y-intercept. [3]

(b) For the graph of $y = f(x)$, find the equation of:

(i) the horizontal asymptote;

(ii) the vertical asymptote. [3]

(c) Hence sketch the graph of $y = f(x)$. [2]

Working:

Question 21 Source worksheet · p.31

Question 21 Non-calculator SL & HL Medium [Maximum mark: 5]

Source: extra practice question

The function $h(x) = \dfrac{ax+b}{x+c}$ has a vertical asymptote $x = 2$ and a horizontal asymptote $y = 3$. The graph of h passes through the point $(0, 1)$.

Find the values of a, b and c. [5]

Working:

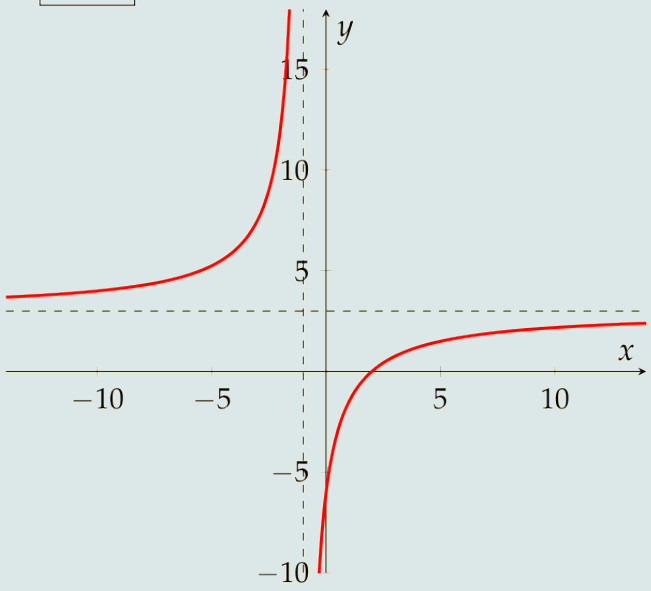
Answer 19 Source worksheet · p.27

Question 19

(a)(i) $3x - 6 = 0$: $\boxed{(2,\ 0)}$ (ii) $f(0) = -6$: $\boxed{(0,\ -6)}$

(b)(i) $\boxed{y = 3}$ (ii) $\boxed{x = -1}$

(c)



28

Answer 21 Source worksheet · p.31

Question 21

Vertical asymptote where $x + c = 0$: $-c = 2 \Rightarrow \boxed{c = -2}$.

Horizontal asymptote $y = \frac{a}{1}$: $\boxed{a = 3}$.

Through $(0, 1)$: $\frac{b}{-2} = 1 \Rightarrow \boxed{b = -2}$. So $h(x) = \frac{3x - 2}{x - 2}$.

Question 8

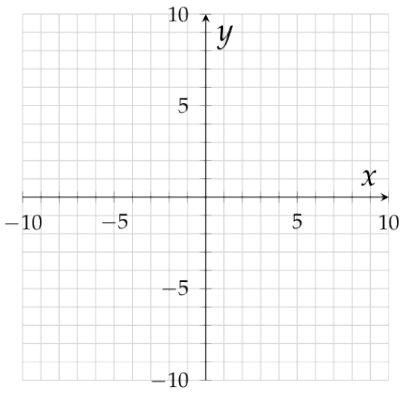
Source worksheet · p.15

Question 8

Non-calculator HL only Medium [Maximum mark: 7]

Sketch $f(x) = \dfrac{1}{x^2 - 4}$. Include a sign diagram.

Working:



Question 9

Source worksheet · p.17

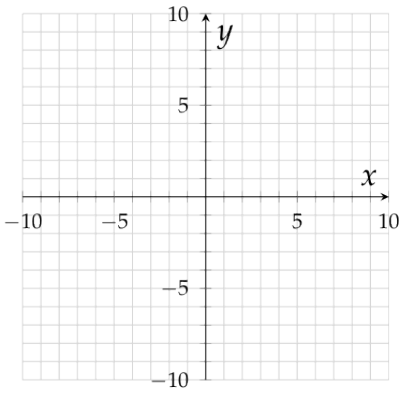
Question 9

Non-calculator HL only Medium [Maximum mark: 7]

Sketch $f(x) = \dfrac{2}{x^2 - 2x - 3}$. Include a sign diagram.

Hint: use the "reciprocal of the parabola" tip from Worked example 3 to find the turning point.

Working:



Answer 8 Source worksheet · p.15

Question 8

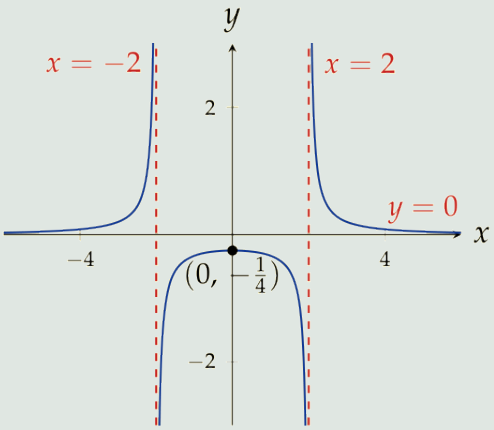
$$f(x) = \frac{1}{(x+2)(x-2)}$$

x-int: **none** (numerator $1 \neq 0$)

y-int: $-\frac{1}{4} \Rightarrow \left(0, -\frac{1}{4}\right)$ (a maximum)

V.A.: $\boxed{x=-2}$, $\boxed{x=2}$ H.A.: $\boxed{y=0}$

	-2		2	
$-$	0	$+$		$+$ $(x+2)$
$-$		$-$	0	$+$ $(x-2)$
$+$	‖	$-$	‖	$+$ $f(x)$



Range: $y > 0$ or $y \leq -\frac{1}{4}$

Answer 9 Source worksheet · p.17

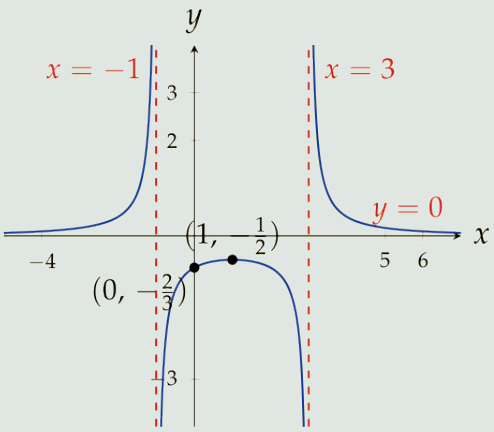
Question 9

$$f(x) = \frac{2}{(x-3)(x+1)}$$

x-int: **none** y-int: $\frac{2}{-3} \Rightarrow \left(0, -\frac{2}{3}\right)$

V.A.: $\boxed{x=-1}$, $\boxed{x=3}$ H.A.: $\boxed{y=0}$

	-1		3	
$-$	0	$+$		$+$ $(x+1)$
$-$		$-$	0	$+$ $(x-3)$
$+$	‖	$-$	‖	$+$ $f(x)$



Turning point: $q = x^2 - 2x - 3$ has its minimum at $x = 1$ (halfway between -1 and 3): $q(1) = -4$, so $f(1) = \frac{2}{-4} = -\frac{1}{2}$. Maximum $\left(1, -\frac{1}{2}\right)$.

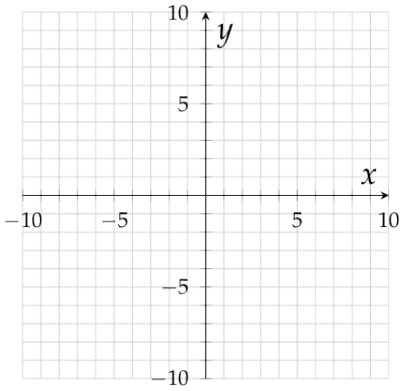
Range: $y > 0$ or $y \leq -\frac{1}{2}$

Question 10 Source worksheet · p.17

Question 10 Non-calculator HL only Medium [Maximum mark: 7]

Sketch $f(x) = \dfrac{x-1}{x^2+x-6}$. Include a sign diagram.

Working:



Question 22 Source worksheet · p.31

Question 22 Non-calculator SL & HL Medium [Maximum mark: 10]

Source: Revision Village (Rational & Inverse Functions Q16)

Consider the functions f and g such that $f(x) = x^2 - 5x + 4$ and $g(x) = \dfrac{x-2}{f(x)}$.

(a) Evaluate $g(3)$. [2]

For the graph of g,

(b) Determine the equations of the vertical asymptotes. [2]

(c) Find the coordinates of any axes intercepts. [2]

(d) Determine the equation of the horizontal asymptote. [1]

(e) Hence sketch the graph of g, labelling any asymptotes and any axes intercepts. [3]

Working:

Answer 10 Source worksheet · p.17

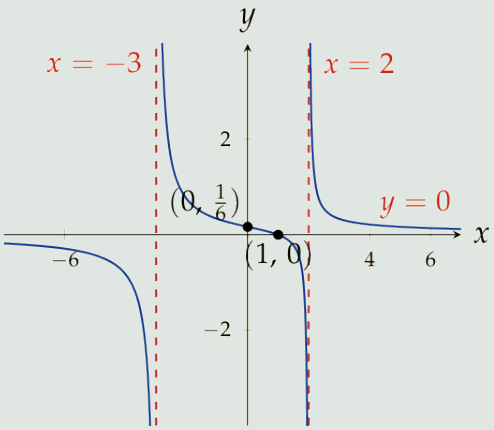
Question 10

$$f(x) = \frac{x-1}{(x+3)(x-2)}$$

x-int: $(1,\,0)$ $\quad$ y-int: $\frac{-1}{-6} \Rightarrow \left(0,\,\frac{1}{6}\right)$

V.A.: $\boxed{x=-3}$, $\boxed{x=2}$ $\quad$ H.A.: $\boxed{y=0}$

	-3		1		2	
$(x-1)$	$-$	$-$	0	$+$	$+$	
$(x+3)$	$-$	0	$+$	$+$	$+$	
$(x-2)$	$-$	$-$	$-$	0	$+$	
$f(x)$	$-$	‖	$+$	0	$-$	‖ $+$



Answer 22 Source worksheet · p.31

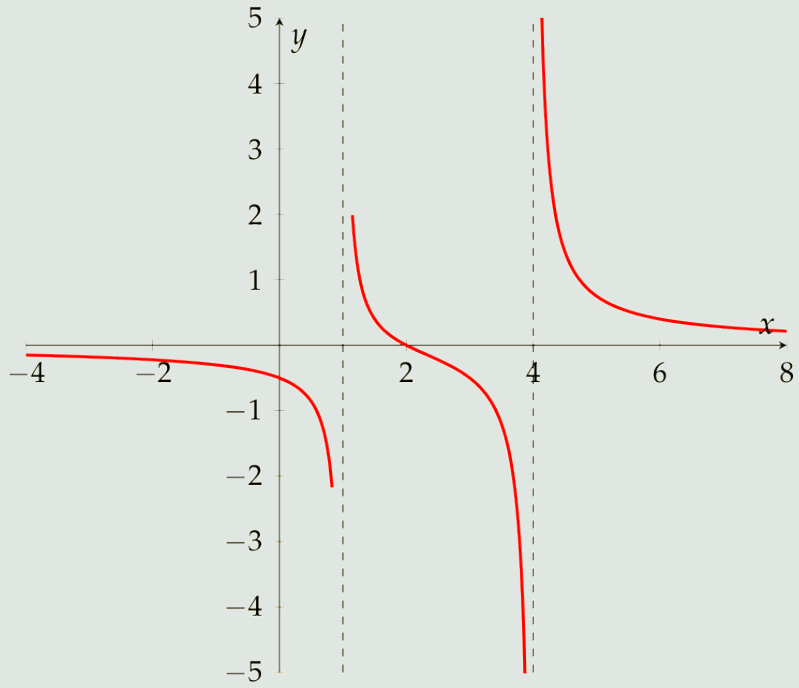
Question 22

$f(x) = (x-1)(x-4)$, so $g(x) = \dfrac{x-2}{(x-1)(x-4)}$.

(a) $g(3) = \dfrac{1}{(2)(-1)} = \boxed{-\frac{1}{2}}$ $\quad$ (b) $\boxed{x=1}$ and $\boxed{x=4}$

(c) x-intercept $\boxed{(2,\,0)}$; y-intercept $g(0) = \dfrac{-2}{4}$: $\boxed{\left(0,\,-\frac{1}{2}\right)}$ $\quad$ (d) degree of top $<$ degree of bottom: $\boxed{y=0}$

(e)



Question 11

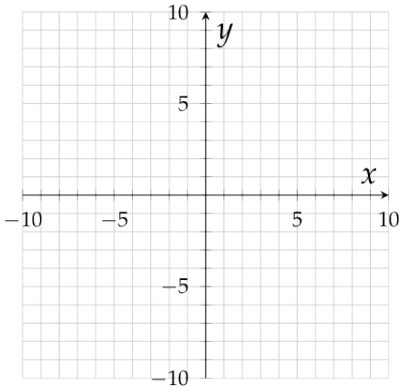
Source worksheet · p.19

Question 11

Non-calculator HL only Medium [Maximum mark: 7]

Let $f(x) = \dfrac{x^2 + 2x - 3}{x + 1}$. Write $f(x)$ in the form $ax + b + \dfrac{c}{x+1}$, write down the asymptotes, and sketch the graph showing the intercepts.

Working:



Question 12

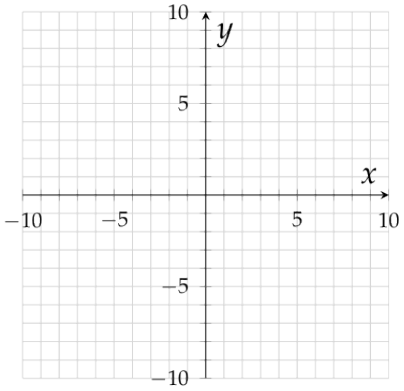
Source worksheet · p.19

Question 12

Non-calculator HL only Medium [Maximum mark: 7]

Let $f(x) = \dfrac{x^2 - 4}{x + 1}$. Find the asymptotes and the intercepts, and sketch the graph.

Working:



Answer 11 Source worksheet · p.19

Question 11

$$\begin{array}{c|ccc} -1 & 1 & 2 & -3 \\ & & -1 & -1 \\ \hline & 1 & 1 & \boxed{-4} \end{array}$$

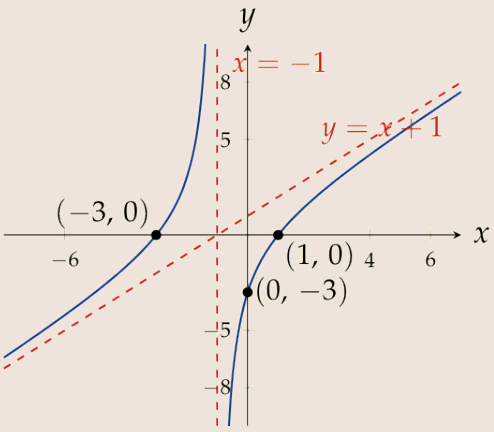
$$\Rightarrow \boxed{f(x) = x + 1 - \frac{4}{x+1}}$$

V.A.: $\boxed{x = -1}$ oblique: $\boxed{y = x + 1}$

x-int: $(x+3)(x-1) = 0 \Rightarrow (-3, 0), (1, 0)$

y-int: $\frac{-3}{1} \Rightarrow (0, -3)$

No turning points – each branch is always increasing.



Answer 12 Source worksheet · p.19

Question 12

$$\begin{array}{c|ccc} -1 & 1 & 0 & -4 \\ & & -1 & 1 \\ \hline & 1 & -1 & \boxed{-3} \end{array}$$

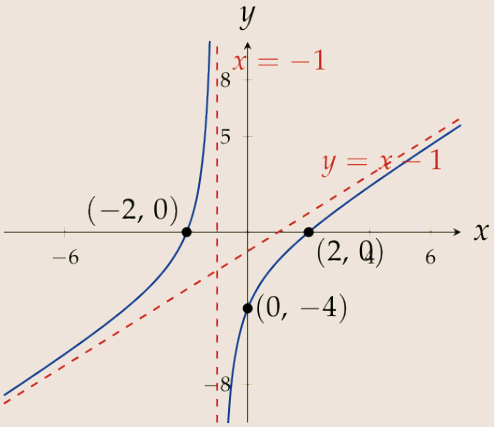
$$\Rightarrow f(x) = x - 1 - \frac{3}{x+1}$$

V.A.: $\boxed{x = -1}$ oblique: $\boxed{y = x - 1}$

x-int: $(x-2)(x+2) = 0 \Rightarrow (-2, 0), (2, 0)$

y-int: $\frac{-4}{1} \Rightarrow (0, -4)$

No turning points – each branch is always increasing.



Question 16

Source worksheet · p.23

Question 16

Calculator HL only Hard

[Maximum mark: 8]

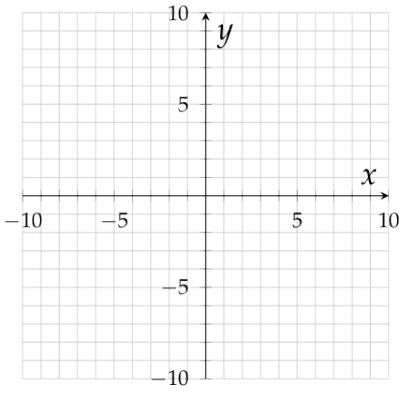
Let $f(x) = \dfrac{x^2 + 3x + 4}{x + 1}$.

(a) Write $f(x)$ in the form $ax + b + \dfrac{c}{x+1}$.

(b) Write down the equations of the asymptotes.

(c) Sketch the graph, showing any intercepts and turning points.

Working:



Answer 16 Source worksheet · p.23

Question 16

(a)

-1	1	3	4
		-1	-2
	1	2	$\boxed{2}$

$$\boxed{f(x) = x + 2 + \frac{2}{x+1}}$$

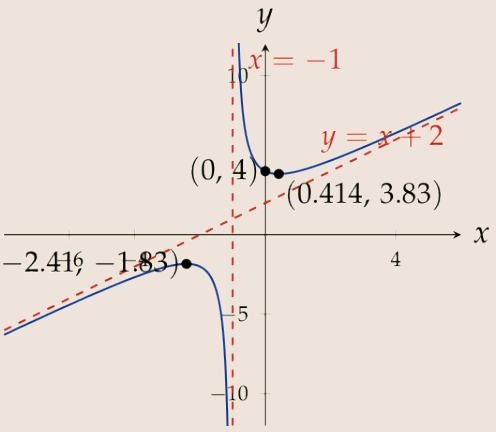
(b) V.A.: $\boxed{x = -1}$, oblique: $\boxed{y = x + 2}$

(c) x-int: $\Delta = 9 - 16 < 0$ – **none**

y-int: $(0, 4)$

max $(-2.41, -1.83)$

min $(0.414, 3.83)$



Question 17

Source worksheet · p.25

Question 17

Calculator HL only Hard

[Maximum mark: 9]

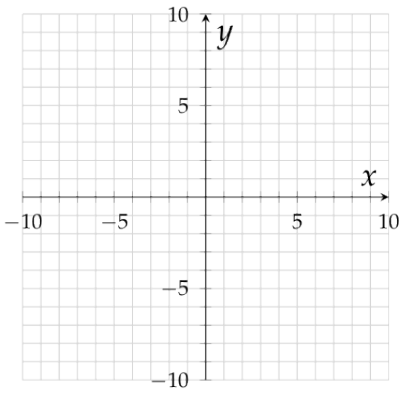
Let $f(x) = \dfrac{2x^2 + x - 3}{x - 2}$.

(a) Find the equation of the oblique asymptote.

(b) Find the x- and y-intercepts.

(c) Sketch the graph, showing the asymptotes, intercepts and turning points.

Working:



Answer 17 Source worksheet · p.25

Question 17

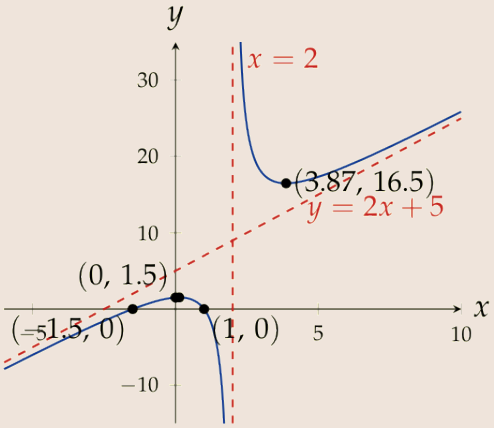
(a)

2	2	1	-3
		4	10
	2	5	$\boxed{7}$

$$f(x) = 2x + 5 + \frac{7}{x-2}$$

Oblique asymptote: $\boxed{y = 2x + 5}$

(V.A. $x = 2$)



(b) $2x^2 + x - 3 = (2x+3)(x-1) = 0$

x-int: $(-1.5, 0)$ and $(1, 0)$

y-int: $\frac{-3}{-2} \Rightarrow (0, 1.5)$

(c) max $(0.129, 1.52)$, min $(3.87, 16.5)$

End of notes

Question 25 Source worksheet · p.37

Question 25

Non-calculator | HL only | Hard

[Maximum mark: 17]

Source: Revision Village (Rational & Inverse Functions Q15)

Consider the function f such that $f(x) = \dfrac{2x^2 - 10x + 20}{x - 3}$.

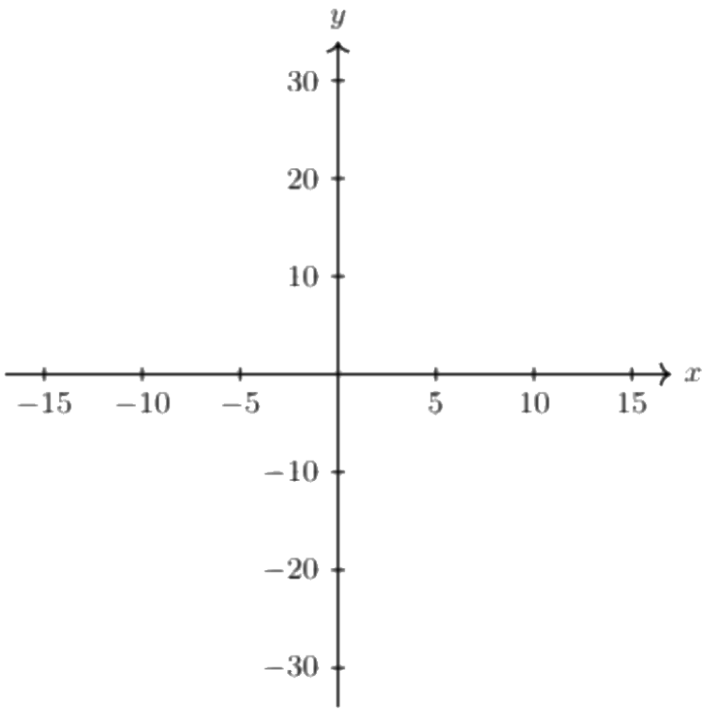
(a) Write down the equation of the vertical asymptote. [1]

(b) Find the coordinates of any maxima and minima on the graph of f. [6]

The graph of f approaches an oblique asymptote as $x \to \pm\infty$.

(c) Find the equation of the oblique asymptote. [4]

(d) Sketch the graph of f on the axes provided, clearly indicating any asymptotes, axes intercepts, maxima and minima. [5]



(e) State the range of f. [1]

Working:

Answer 25 Source worksheet · p.37

Question 25

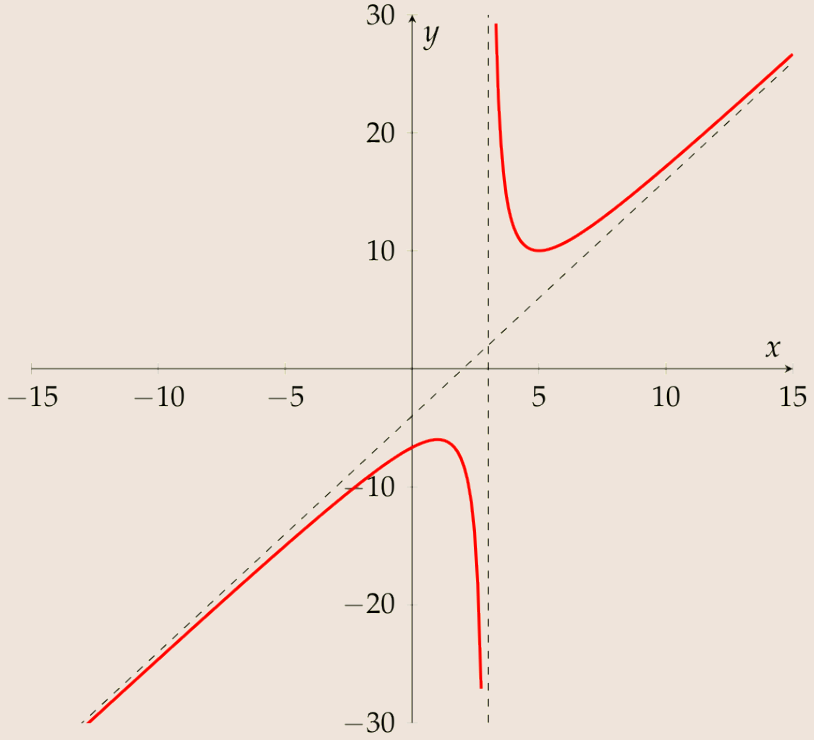
(a) $\boxed{x = 3}$

(b) $f'(x) = \dfrac{(4x-10)(x-3)-(2x^2-10x+20)}{(x-3)^2} = \dfrac{2x^2-12x+10}{(x-3)^2} = \dfrac{2(x-1)(x-5)}{(x-3)^2}$

$f(1) = -6$, $f(5) = 10$: maximum $\boxed{(1, -6)}$, minimum $\boxed{(5, 10)}$

(c) $2x^2 - 10x + 20 = (x-3)(2x-4) + 8$, so $f(x) = 2x - 4 + \dfrac{8}{x-3}$: $\boxed{y = 2x - 4}$

(d) y-intercept $\left(0, -\frac{20}{3}\right)$; no x-intercepts ($b^2 - 4ac = 100 - 160 < 0$).



(e) $\boxed{f(x) \leq -6 \text{ or } f(x) \geq 10}$

Question 13

Source worksheet · p.21

Question 13

Calculator SL & HL Medium [Maximum mark: 8]

Let $f(x) = \dfrac{2x+1}{x-3}$.

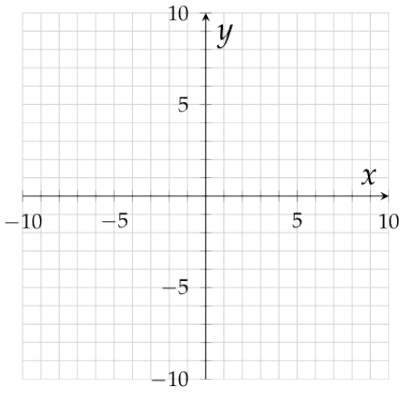
(a) Write down the equations of the asymptotes.

(b) Find the intercepts.

(c) Sketch the graph of f and the line $y = x$ on the same axes.

(d) Use your GDC to find the coordinates of the points where the graph of f meets the line $y = x$.

Working:



Question 14

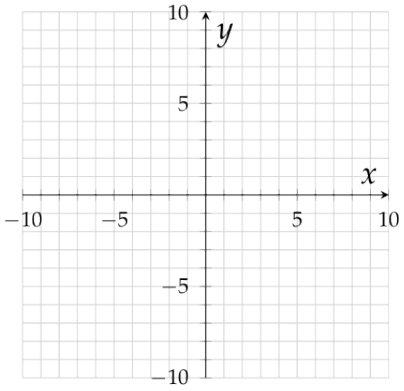
Source worksheet · p.21

Question 14

Calculator HL only Medium [Maximum mark: 7]

Sketch $f(x) = \dfrac{4x}{x^2+4}$, and state the range. (Use your GDC for any turning points.)

Working:



Answer 13 Source worksheet · p.21

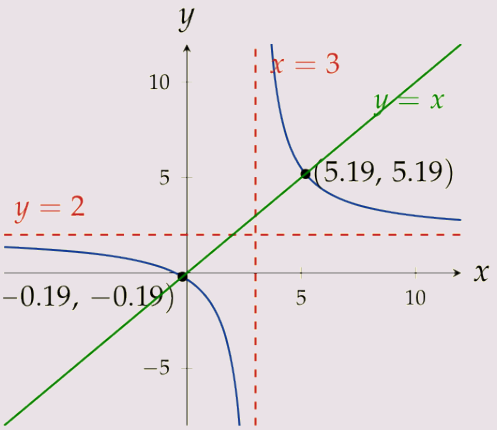
Question 13

(a) V.A.: $\boxed{x = 3}$ H.A.: $\boxed{y = 2}$

(b) x-int: $\left(-\frac{1}{2}, 0\right)$ y-int: $\left(0, -\frac{1}{3}\right)$

(c) See the graph.

(d) Solve $\dfrac{2x+1}{x-3} = x$ on the GDC (or $x^2 - 5x - 1 = 0$):
$\boxed{(5.19, 5.19)}$ and $\boxed{(-0.193, -0.193)}$



Answer 14 Source worksheet · p.21

Question 14

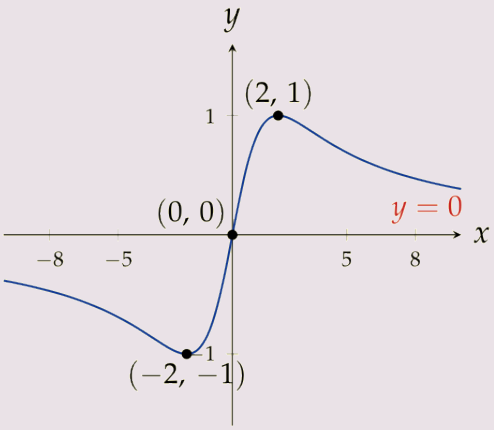
$x^2 + 4 = 0$ has **no** real solutions, so there is **no vertical asymptote**. Domain: $x \in \mathbb{R}$.

Intercept: $(0, 0)$ H.A.: N<D $\Rightarrow$ $\boxed{y = 0}$

Turning points (GDC): max $(2, 1)$, min $(-2, -1)$

	0		
−	0	+	$4x$
+		+	$x^2 + 4$
−	0	+	$f(x)$

Range: $\boxed{-1 \leq y \leq 1}$



Question 15

Source worksheet · p.23

Question 15

Calculator HL only Hard [Maximum mark: 7]

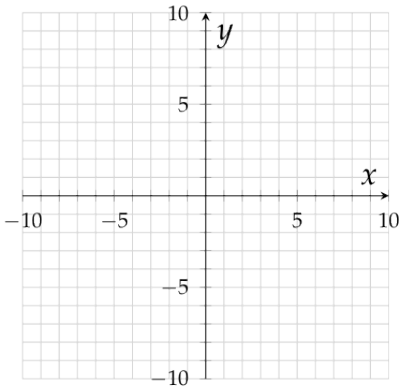
Let $f(x) = \dfrac{3x+1}{x^2+1}$.

(a) Explain why the graph has no vertical asymptote, and write down the horizontal asymptote.

(b) Find the intercepts.

(c) Use your GDC to find the turning points, and hence sketch the graph and state the range.

Working:



Answer 15 Source worksheet · p.23

Question 15

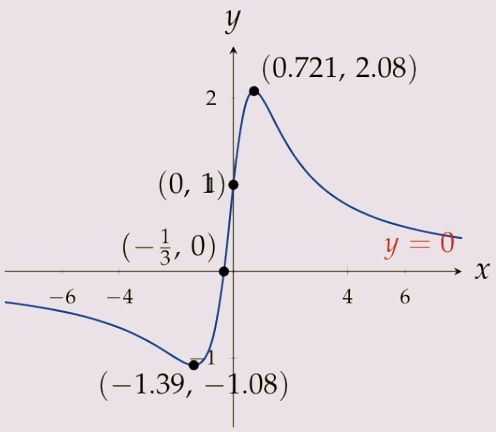
(a) $x^2 + 1 > 0$ for all x, so the denominator is never 0 – **no V.A.**

N<D $\Rightarrow$ H.A. $\boxed{y = 0}$

(b) x-int: $\left(-\frac{1}{3}, 0\right)$ y-int: $(0, 1)$

(c) max $(0.721, 2.08)$, min $(-1.39, -1.08)$

Range: $\boxed{-1.08 \leq y \leq 2.08}$



Question 20 Source worksheet · p.29

Question 20

Non-calculator | SL & HL | Easy

[Maximum mark: 7]

Source: Revision Village (Rational & Inverse Functions Q6)

A function f is defined by $f(x) = \dfrac{a - 3x}{x - 2}$, where $x \in \mathbb{R}$, $x \neq 2$ and $a \in \mathbb{Z}$.

(a) For the graph of $y = f(x)$, write down the equation of

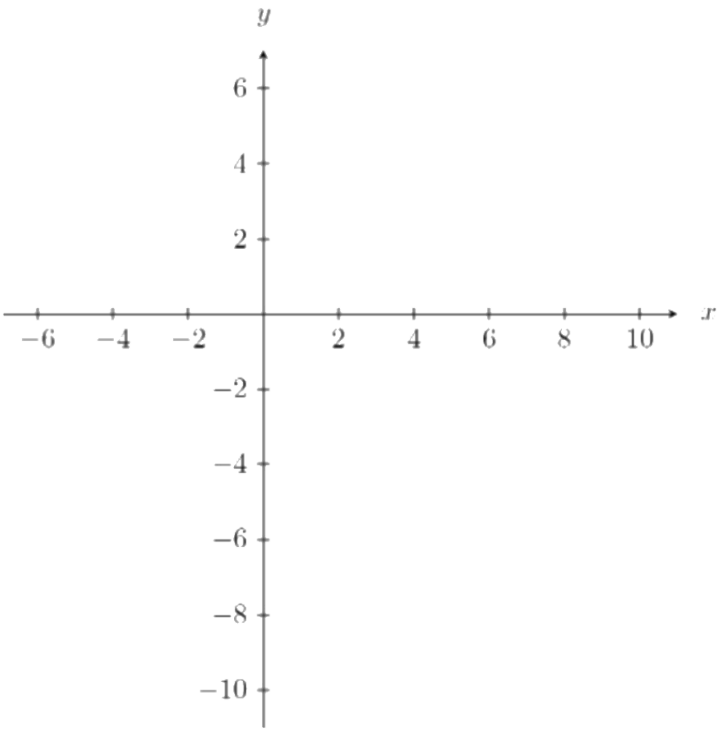
(i) the vertical asymptote;

(ii) the horizontal asymptote. [2]

It is known that $f(1) = f^{-1}(1)$.

(b) Find the value of a. [3]

(c) Hence, sketch the graph of f on the axes below. On your sketch, indicate the values of any axis intercepts and label any asymptotes with their equations.



[2]

Working:

Answer 20

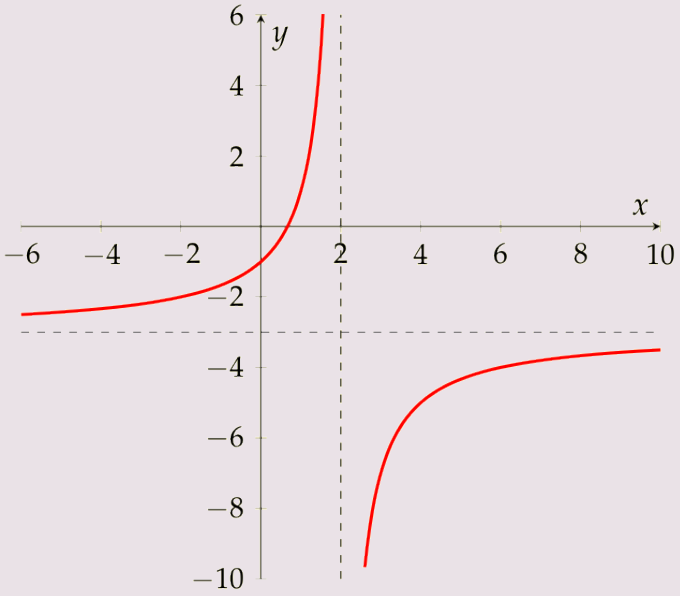
Source worksheet · p.29

Question 20

(a)(i) $\boxed{x = 2}$ (ii) $\boxed{y = \quad 3}$

$= 1 \Rightarrow \boxed{a = 2}$

(c) $f(x) = \dfrac{\quad}{x-2}$: y-intercept $(0, -1)$, x-intercept $\left(\frac{2}{3}, 0\right)$



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Checked proof: $f(1) = 3 - a$ and $f^{-1}(1) = \frac{a+2}{4}$. Thus $3 - a = \frac{a+2}{4}$ gives $a = 2$; no universal inverse-intersection claim is needed.

Question 23 Source worksheet · p.33

Question 23 Calculator SL & HL Easy [Maximum mark: 7]

Source: Revision Village (Rational & Inverse Functions Q8)

The function f is given by $f(x) = \dfrac{4x^2 - 128}{x^2 - 16}$, for $x \neq \pm 4$.

(a) Prove that f is an even function. [2]

(b) (i) Sketch the graph of $y = f(x)$.

(ii) Write down the range of f. [5]

Working:

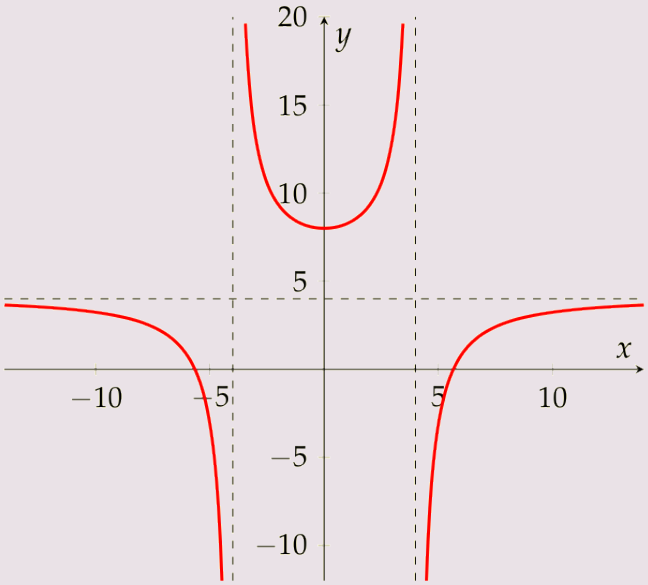
Answer 23

Source worksheet · p.33

Question 23

(a) $f(-x) = \dfrac{4(-x)^2 - 128}{(-x)^2 - 16} = \dfrac{4x^2 - 128}{x^2 - 16} = f(x)$, so f is even (symmetric about the y-axis).

(b)(i) Asymptotes $x = \pm 4$, $y = 4$; $(0, 8)$ is a minimum; x-intercepts $(\pm\sqrt{32}, 0) \approx (\pm 5.66, 0)$.



(ii) $\boxed{y < 4 \text{ or } y \geq 8}$

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Question 24 Source worksheet · p.35

Question 24 Calculator SL & HL Medium [Maximum mark: 17]

Source: Revision Village (Rational & Inverse Functions Q12)

Let $f(x) = \dfrac{3x+4}{2x-5}$, for $x \neq \dfrac{5}{2}$.

(a) For the graph of $y = f(x)$, find the coordinates of:

(i) the x-intercept;

(ii) the y-intercept. [3]

(b) For the graph of $y = f(x)$, find the equation of:

(i) the horizontal asymptote;

(ii) the vertical asymptote. [3]

(c) Hence sketch the graph of $y = f(x)$. [2]

(d) Find $f^{-1}(x)$. [3]

The graphs of $f(x)$ and $f^{-1}(x)$ have two points of intersection.

(e) Find the coordinates of the two points of intersection. **Give your answers to two decimal places**. [3]

(f) Hence find the equation of the perpendicular bisector to the two points of intersection. [3]

Working:

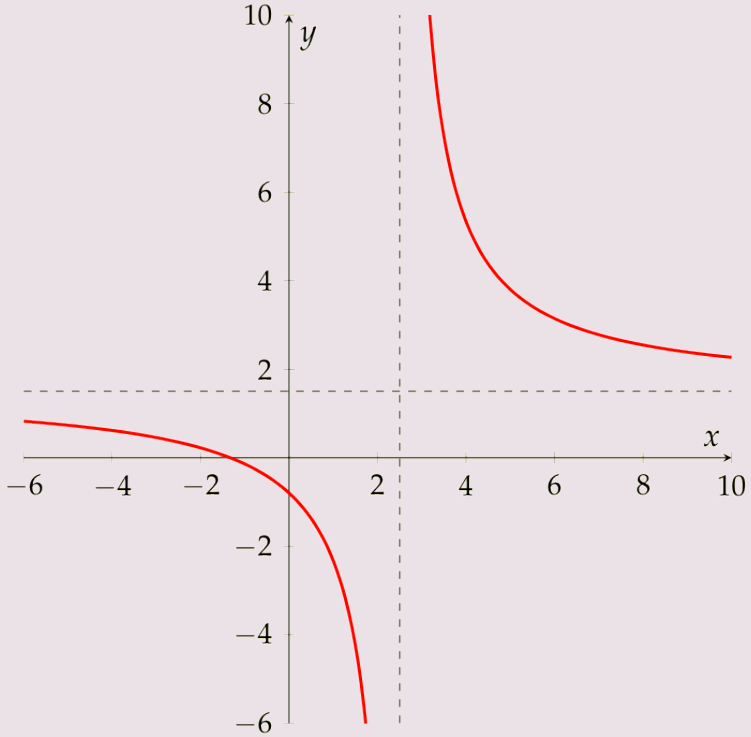
Answer 24 Source worksheet · p.35

Question 24

(a)(i) $3x + 4 = 0$: $\boxed{\left(-\frac{4}{3}, 0\right)}$ (ii) $f(0)$: $\boxed{\left(0, -\frac{4}{5}\right)}$

(b)(i) $\boxed{y = \frac{3}{2}}$ (ii) $\boxed{x = \frac{5}{2}}$

(c)



(d) $x = \dfrac{3y+4}{2y-5} \Rightarrow 2xy - 5x = 3y + 4 \Rightarrow y(2x-3) = 5x + 4$: $\boxed{f^{-1}(x) = \dfrac{5x+4}{2x-3}}$

(e) Intersections lie on $y = x$: $2x^2 - 8x - 4 = 0 \Rightarrow x = 2 \pm \sqrt{6}$: $\boxed{(4.45, 4.45)}$ and $\boxed{(-0.45, -0.45)}$

(f) Midpoint $(2, 2)$, gradient of chord 1, so perpendicular gradient -1: $\boxed{y = -x + 4}$